scRNA-seq Analysis Report
260504-pbmc3k

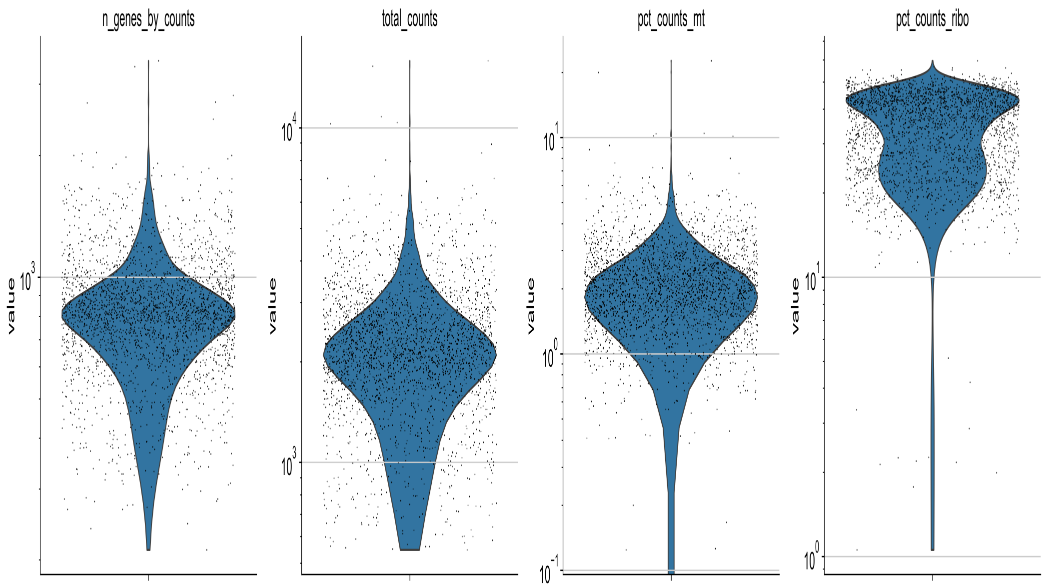
Project name: 260504-pbmc3k
Tissue / dataset type: N/A
Species: N/A
Date: 2026-05-06
Results directory: K:\@scRNA-1.1-PBMC3K\results\260504-pbmc3k
Figures directory: K:\@scRNA-1.1-PBMC3K\figures\260504-pbmc3k

1. Pipeline Overview

This report summarizes a standardized single-cell RNA-seq analysis pipeline, including: quality control, normalization and log-transformation, dimensionality reduction (PCA and UMAP), graph-based clustering (Leiden), marker-gene analysis, automatic cell-type annotation based on reference markers, and supplementary evaluation of annotation confidence and cluster consistency.

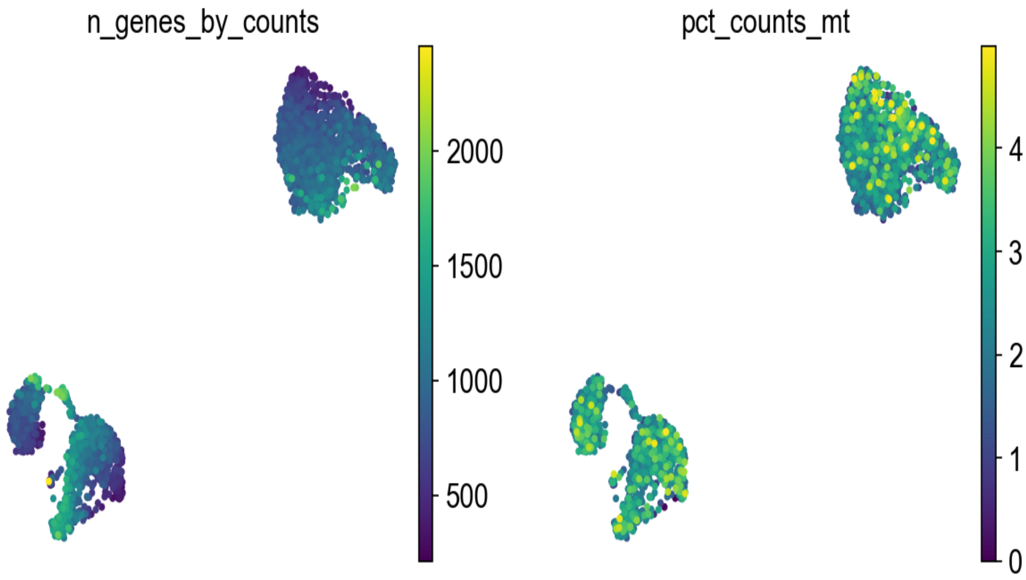
2. Quality Control (QC)

QC parameters (from YAML)
min_genes_per_cell: 200
max_genes_per_cell: 2500
max_pct_mt: 5.0
min_cells_per_gene: 3



3. UMAP Embeddings & Clustering

UMAP embeddings provide a 2D representation of the high-dimensional transcriptomic space. Cells are colored by QC status, clusters, or cell types to illustrate global structure and major cell populations.



4. Cell-Type Annotation & Marker Genes

Automatic cell-type annotation was performed using 9 reference marker sets defined in the YAML configuration. Each cell was assigned the label whose marker gene score was highest.

cluster	gene	logFC	pval_adj
0	LDHB	2.77	2.3e-231
0	RPS12	1.04	1.0e-221
0	RPS25	1.17	6.0e-195
1	CD74	4.09	2.1e-181
1	CD79A	7.73	2.0e-165
1	HLA-DRA	4.89	1.4e-164
2	LYZ	6.20	7.4e-240
2	S100A9	7.39	8.2e-240
2	S100A8	7.56	5.7e-231
3	NKG7	7.11	3.5e-215
3	CST7	6.28	4.3e-162
3	GZMA	6.14	2.3e-159
4	SAT1	3.67	3.1e-84
4	COTL1	3.47	7.2e-84
4	FCER1G	4.52	5.1e-83
5	HLA-DPB1	4.63	4.1e-16
5	HLA-DPA1	4.42	4.1e-16
5	HLA-DRB1	4.45	1.4e-15

6. Key Analysis Parameters (from YAML)

QC
min_genes_per_cell: 200

max_genes_per_cell: 2500
max_pct_mt: 5.0
min_cells_per_gene: 3

Analysis
n_top_hvgs: 2000
n_pcs: 40
neighbor_k: 15
leiden_resolution: 0.5

Annotation
reference marker groups: 9

Total cells: 2638

Total genes: 2000